# Rank-27 Current Broad-Search Call-Graph Closure

Tactix Development Project

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## Outcome

A fresh current-source call graph reproduces the exact accepted Rank-27 conflict-9450 state and leaves zero residue, but it does not identify a new portable worse-side repair. Its leading samples belong to mechanisms already closed by stronger normalized evidence or to components that are not slower than their current-Kissat counterparts. No solver behavior changes in this increment.

## 1 Immutable observation

Single-solver job `1650249` ran on node018 with 12 logical CPUs reserved, exactly 48,000 MiB scheduler memory, and a 16 GiB process cap. It completed `0:0` in 15 seconds and stopped intentionally UNKNOWN at conflict 9450. The semantic transcript has SHA-256 `3b87bb36da711bcef3ca877f3922e9fa5f8a3c6ae9d638c1ae18d4965bd92c5f`, exactly the accepted common-state authority. Internal/wall time is 12.828/13.61 seconds, solver maximum RSS is 379,208 KiB, and major page faults are zero.

The immutable RelWithDebInfo pin has SHA-256 `3cd454dee1942a79b23544ace94f988526d9edca9082950bbf420bca590bed87`. Current Kissat was not rerun; the accepted exact-boundary profile with SHA-256 `e506ae7f2a3660cd2dc8d8781f8ae1f6c738a0c5eb64e8d2d052c25638652924` remains the source comparator.

## 2 Profile classification

perf records approximately 49.751 billion cycles in 2,665 samples with zero lost samples. The largest self shares are:

| Owner | Self share |
|---|---|
| Ordinary watched propagation | 31.29% |
| Heap bubble-down | 4.35% |
| Substitute SCC discovery | 3.33% |
| Collection | 3.30% |
| Binary-live queries | 3.02% |
| Unassignment | 2.28% |
| Analysis application | 2.11% |
| Backtrack | 1.97% |

These percentages do not reopen their mechanisms. Propagation and active selection have normalized current-Kissat comparisons; the first heap choice after the common boundary is an exact

equal-score topology tie; collection's remaining owner is call frequency with its known alternatives closed; and the Substitute, Vivify, residual BVE, and Congruence candidates have exact current dispositions. In particular, the earlier exact Congruence substage observer priced Tactix binary-liveness construction at 85 ms while current Kissat's broader dense-entry, binary-extraction, and dense-exit preparation cost 198 ms. The sampled binary-live share is therefore not contradictory source-equivalent evidence for another Congruence layout.

## 3 Next localization

Broad-search convergence now follows semantic quality past recorded decision 5,795,255 without repairing its neutral equal-score tie. The next observer must name the first unequal-score or candidate-set divergence, or the first point where comparable cost leaves Tactix with weaker learning, pruning, simplification, or retained state. Exact scheduling, tick/cost, candidate order, learning, and pruning remain required while the states are common.

The staged CNF and node-local scratch are absent. The bounded run created no proof, model, checker, certificate, DRAT, or LRAT artifact.